

module4_lab

September 23, 2026

1 Module 4 Lab 1: Describe and Correlate

Repo URL: <https://github.com/roee0910/cs82a-portfolio.git>

Data: my own sales_clean.csv from Lab 3.

```
[1]: import pandas as pd
import matplotlib.pyplot as plt
```

1.1 Step 1: Load your clean data

```
[2]: df = pd.read_csv("sales_clean.csv")
print(df.shape)
df.head()
```

(292, 6)

```
[2]:   order_id      date  product  price  qty  zip
0      1254  2026-04-06      mug   11.36   4  60614
1      1057  2026-05-30  charger   14.79   3  30303
2      1150  2026-05-06      mug    5.50   1  10001
3      1066  2026-05-30  notebook   37.53   1  98101
4      1114  2026-04-06  webcam   45.59   4  90405
```

1.2 Step 2: Center and spread of price and qty

```
[3]: df[["price", "qty"]].agg(["mean", "median", "std"]).round(2)
```

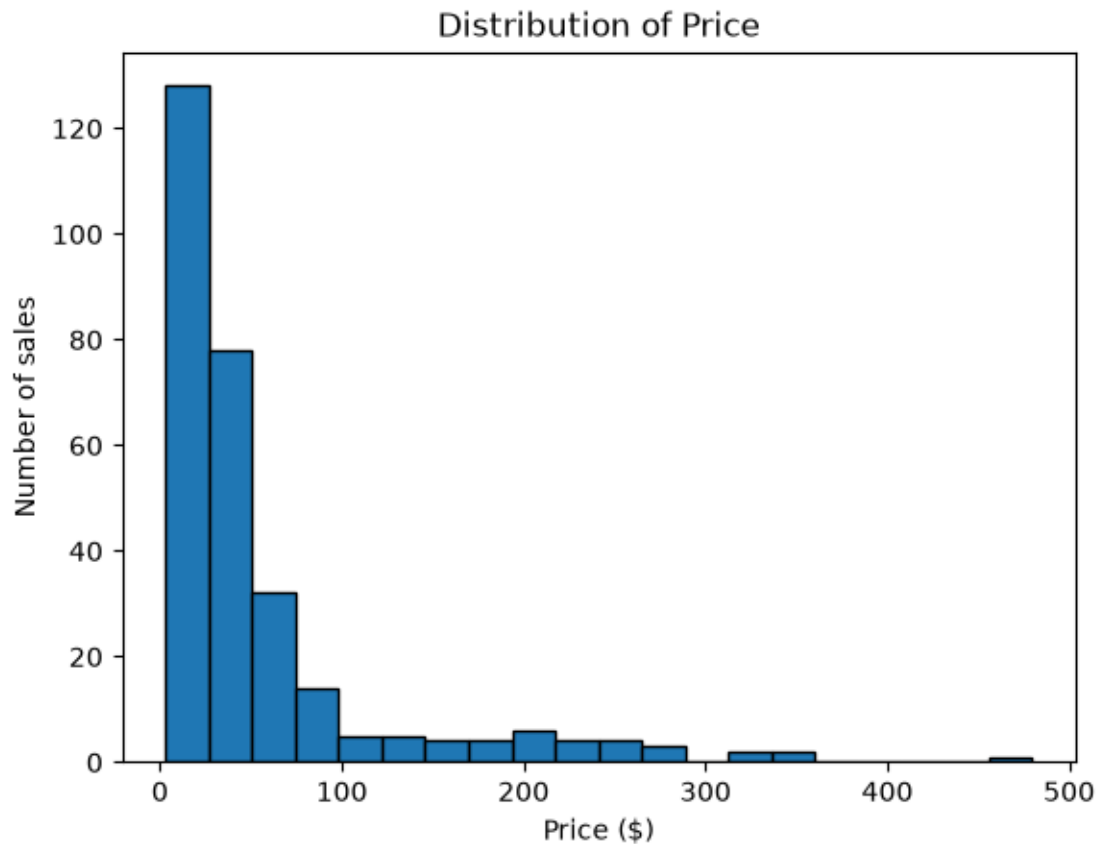
```
[3]:      price  qty
mean    55.52  2.99
median   37.53  3.00
std     71.77  1.41
```

1.3 Step 3: Which summary would I report for price?

I would report the median for price. The mean is \$55.52 but the median is only \$37.53, which tells me that a few expensive items are pulling the average up. The median is closer to what a typical sale looks like.

1.4 Step 4: Histogram of price

```
[4]: df["price"].plot(kind="hist", bins=20, edgecolor="black")
plt.title("Distribution of Price")
plt.xlabel("Price ($)")
plt.ylabel("Number of sales")
plt.show()
```



Shape: The histogram is right-skewed. Most sales are at low prices (the tallest bar is at the far left), and there is a long tail of expensive items going up to about \$480. This matches my choice of the median in Step 3.

1.5 Step 5: Create the revenue column

```
[5]: df["revenue"] = df["price"] * df["qty"]
df.head()
```

```
[5]:   order_id    date  product  price  qty  zip  revenue
0     1254 2026-04-06      mug   11.36    4  60614    45.44
1     1057 2026-05-30  charger   14.79    3  30303    44.37
```

2	1150	2026-05-06	mug	5.50	1	10001	5.50
3	1066	2026-05-30	notebook	37.53	1	98101	37.53
4	1114	2026-04-06	webcam	45.59	4	90405	182.36

1.6 Step 6: Correlation matrix

```
[6]: df.corr(numeric_only=True).round(3)
```

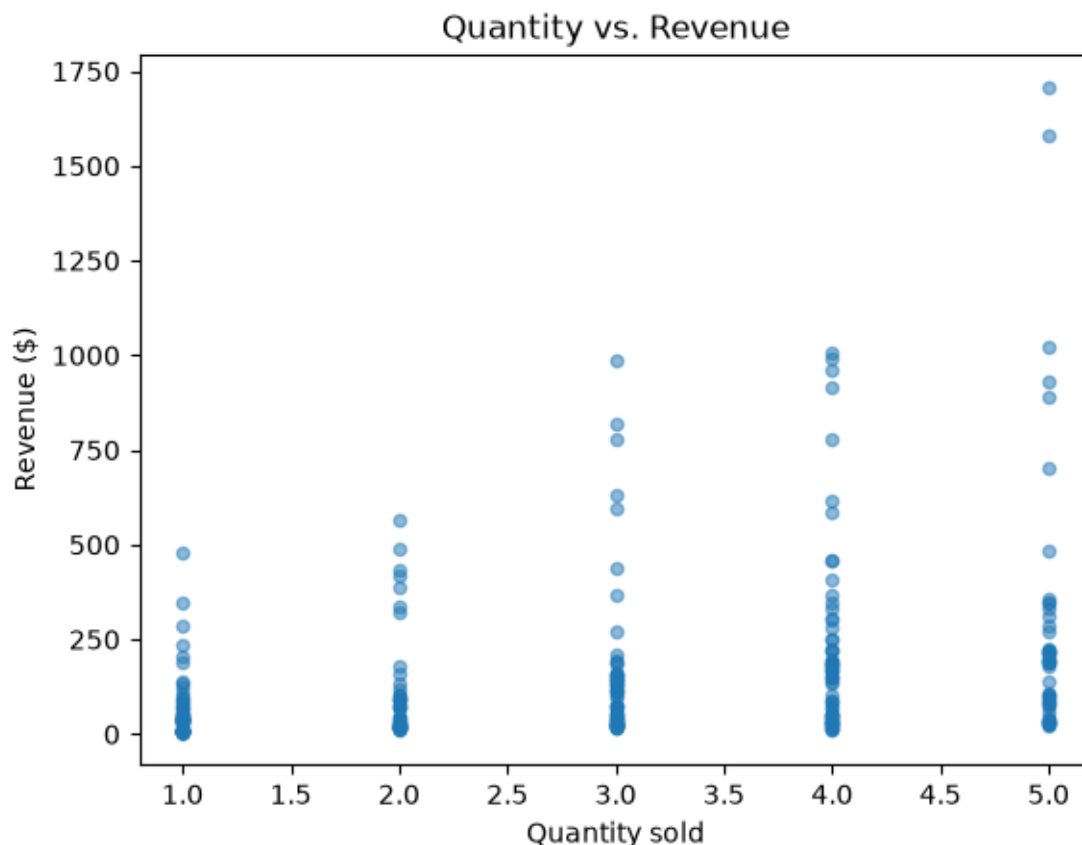
```
[6]:
```

	order_id	price	qty	zip	revenue
order_id	1.000	-0.005	0.031	-0.015	0.004
price	-0.005	1.000	-0.042	0.024	0.821
qty	0.031	-0.042	1.000	0.087	0.313
zip	-0.015	0.024	0.087	1.000	0.047
revenue	0.004	0.821	0.313	0.047	1.000

The strongest off-diagonal value is price vs. revenue ($r = 0.82$), but that one is partly built in, since revenue is price $\times$ qty. The pair I care about is qty vs. revenue, with $r = 0.31$, a weak to moderate positive relationship. Price and qty are almost unrelated ($r = -0.04$).

1.7 Step 7: Scatter plot to verify

```
[7]: df.plot.scatter(x="qty", y="revenue", alpha=0.5)
plt.title("Quantity vs. Revenue")
plt.xlabel("Quantity sold")
plt.ylabel("Revenue ($)")
plt.show()
```



The scatter shows a mild upward drift, which agrees with $r = 0.31$. Because qty only takes the values 1 to 5, the points sit in vertical columns, and the spread gets wider as qty goes up.

1.8 Step 8: The analyst's sentence

Orders with a higher quantity tend to bring in more revenue, but the link is only weak to moderate ($r = 0.31$), so quantity by itself explains only about 10% of the differences in revenue. Be careful, because price matters more, a few expensive items like monitors create the very large orders, and this correlation does not prove that getting customers to buy more units would raise revenue.

1.9 Step 9: By-hand check

List: [2, 4, 4, 6, 9]

Mean by hand: $2 + 4 + 4 + 6 + 9 = 25$, and $25 / 5$ numbers = **5.0**

Median by hand: the list is already sorted and has 5 numbers, so the median is the middle (3rd) one, which is 4.

Now I verify both with Python:

```
[8]: data = pd.Series([2, 4, 4, 6, 9])  
      print("mean:", data.mean())  
      print("median:", data.median())
```

mean: 5.0

median: 4.0